

Abstract

The Automated Washroom Management System will increase community washroom hygiene, resource consumption, and maintenance response at a minimal cost by automating hygiene and resource use while also using IoT technology. The system is based on an Arduino Uno microcontroller, which is connected to a number of sensors and actuators to provide touchless operation, real-time monitoring, and automatic warnings. The main features include occupancy, automated door control, odour control, flood detection, toilet paper control, and IoT-based maintenance reminders via ThingSpeak and Telegram. The technique includes real-time sensor data collection, embedded control logic decision-making, and cloud-based alarm generation to manage reactive facilities. Experiments have demonstrated that occupancy detection, alarm triggering, and actuator response are reliable. In addition to technical performance, the project prioritises sustainability by promoting energy efficiency, reducing water and paper waste, and improving hygiene in common areas. The technology is scalable and serves as a socially significant example of computer interfacing and embedded internet of things systems being used to improve community infrastructures.

Keywords

ESP, Servo Motor, Automated Odor Management, IoT, Flood Alert system

1. Introduction

Frequently utilised facilities, such as university restrooms, are occasionally plagued by long-term concerns such as poor hygiene, flooding, slow repair, and wasteful use. These challenges impact not only the user experience, but also health and sanitation. Although smart toilet systems have been created in the developed world, they are typically expensive and unsuitable for use in large-scale applications in underdeveloped countries.

This concept was motivated by personal experiences in the BRAC University washrooms, where there was a cleanliness issue. Such difficulties as water overflow when using hand showers, inability to clean on time, and non-contact functionality revealed the need for a smarter and more automated solution that is responsive to local limits.

This project intends to create and implement a low-cost Automated Restroom Management System that will improve hygiene, give a solution for proactive maintenance, and reduce resource waste. The project employs computer interfacing principles by combining multiple analogue and digital sensors with a microcontroller to collect real-time environmental and user data, processing it using embedded control logic, and converting it into physical actions via actuators such as servo motors, LEDs, and buzzers. It exhibits the practical application of communication protocols such as I2C for efficient peripheral interface and UART for module-to-module communication, while IoT-based data exchange connects the system to cloud

platforms for remote monitoring and alerting. These elements demonstrate effective hardware-software integration and embedded system design in a real-world application.

2. Related Work/ Inspiration

The concept of an “Automated Restroom Management System” has been widely adopted in highly technologically advanced countries, particularly Japan. Commercial versions, such as the TOTO Washlet, integrate automatic lid opening, a heated seat, a shower head, and odour removal and cleaning capabilities, demonstrating how embedded technologies and automation may significantly improve hygiene and user experience [1]. Such items, however, are primarily meant for private or luxury use and are not economically viable for large-scale implementation in public or institutional settings, particularly in developing nations.

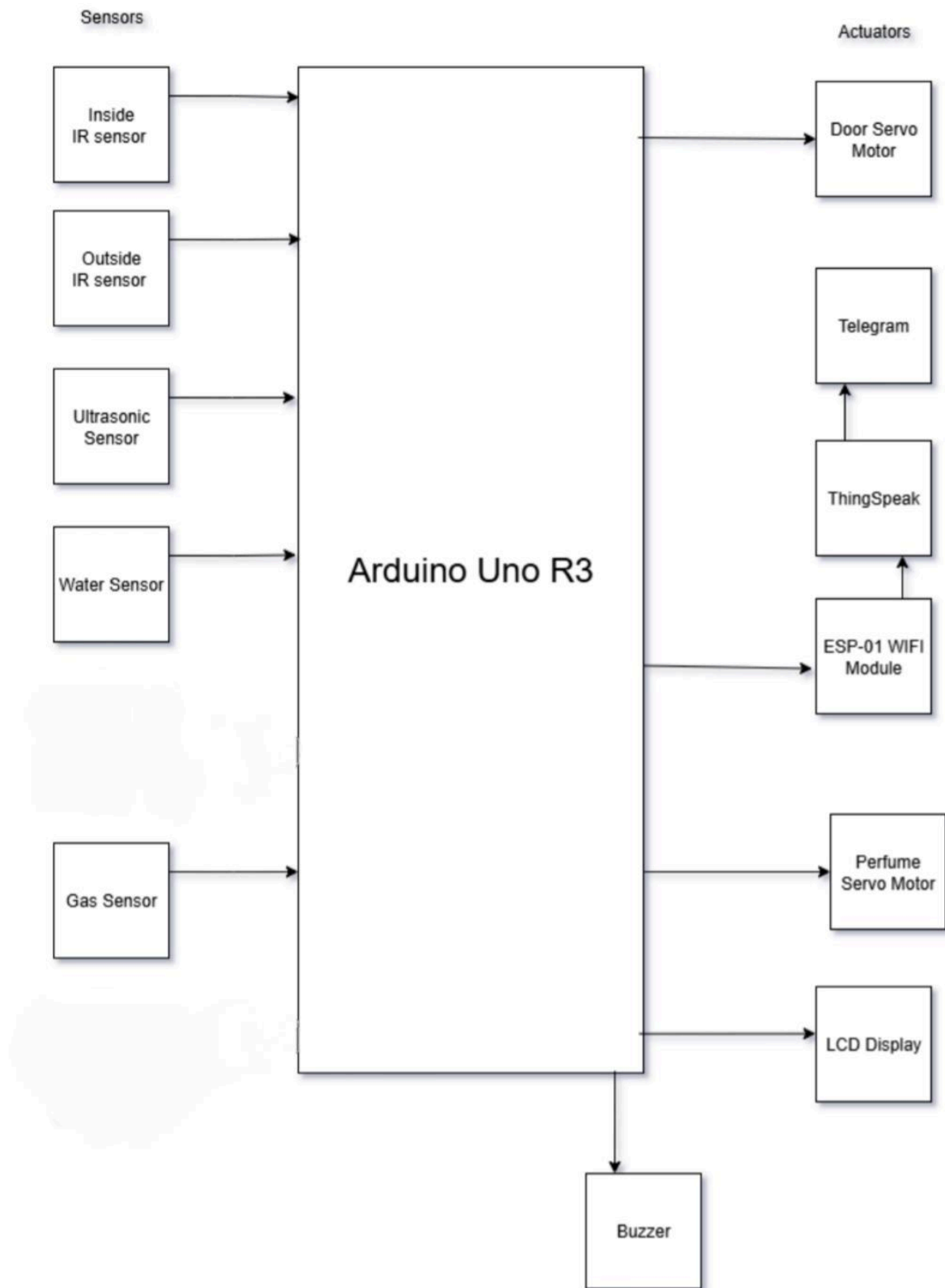
Poor sanitation infrastructure, insufficient supervision, and resource mismanagement are some of the issues faced by Bangladesh's public sanitation services due to low resources. According to studies, poor sanitation facilities have a negative impact on users' health, hygiene, and satisfaction, particularly in congested areas like universities [2]. Commercial smart toilet solutions are typically not resistant to such difficulties, and instead prioritise user convenience at the expense of maintenance and operational control. A variety of scholarly and DIY initiatives have investigated automated sanitation using microcontrollers, sensors, and simple IoT platforms, including occupancy counting, automatic flushing, and air quality monitoring [3][4]. Although the initiatives may be technically possible, they typically lack the real-time alarm systems required for facility operation.

The restrictions outlined in this project are addressed by providing a low-cost, problem-specific smart toilet system that meets BRAC University specifications. The system may transform a typical toilet into a maintenance-conscious gadget by including flood detection, automated resource management, and clean-up notifications via IoT connectivity, resulting in an effective and scalable solution for people to use in institutions.

3. Technical Approach

- **System Architecture:**

The system has a way of working with the Arduino Uno, as the main brain. The Arduino Uno talks to all the sensors, things that do stuff display modules and the WiFi part. The system is always checking what the sensors are saying and looking at the information there while really important things are sent to the cloud so it can make alerts. The Arduino Uno is what makes all this happen. It is always working with the sensors and other parts.



Components Used:

Category	Component	Description
Microcontroller	Arduino Uno R3	The central processing unit for the system
Sensors	HC-SR04 Ultrasonic Sensor	Detects user presence for occupancy status
Sensors	IR Proximity Sensor (x2)	Touchless sensors for opening/closing the door
Sensors	Water Level Sensor	Detects water on the floor for flood alerts
Sensors	MQ-135 Gas Sensor	Detects air quality to trigger the perfume dispenser.
Sensors	5kg Load Cell + HX711 Amplifier	Measures the weight of the toilet paper roll
Actuators	SG90 Servo Motor	Operates the door mechanism
Actuators	MG996R Servo Motor	Operates the perfume dispenser
Actuators	5V Active Buzzer	Provides audible feedback for flushing and alerts
Communication Actuator	ESP-01 WiFi Module	Handles WiFi connectivity and sends IoT alerts
Display & I/O	16×2 I2C LCD Display	Shows system status, sensor data, and alerts
Display & I/O	Red LED & Green LED	Visual indicators for occupied/vacant status
Display & I/O	Push Buttons (x7)	Standalone control (Flush, Cycle Screen, etc.)
Power	4× AA Battery Holder	External power supply for servo motors
Software	Arduino IDE	Used for writing and uploading control code
Software	ThingSpeak	IoT platform to process data and trigger alerts

Software	Telegram	Messaging app for receiving notifications
Basic Components	Breadboard	Holds all components together
Basic Components	10kΩ Resistors	Used to build the resistor ladder for buttons

Cost Breakdown:

Component	Quantity	Estimated Price (BDT)	Total (BDT)
Arduino Uno R3	1	483	483
ESP-01 WiFi Module	1	170	170
ESP-01 Adapter	1	110	110
HC-SR04 Ultrasonic Sensor	1	100	100
IR Sensor	2	45	90
MQ-135 Gas Sensor	1	395	395

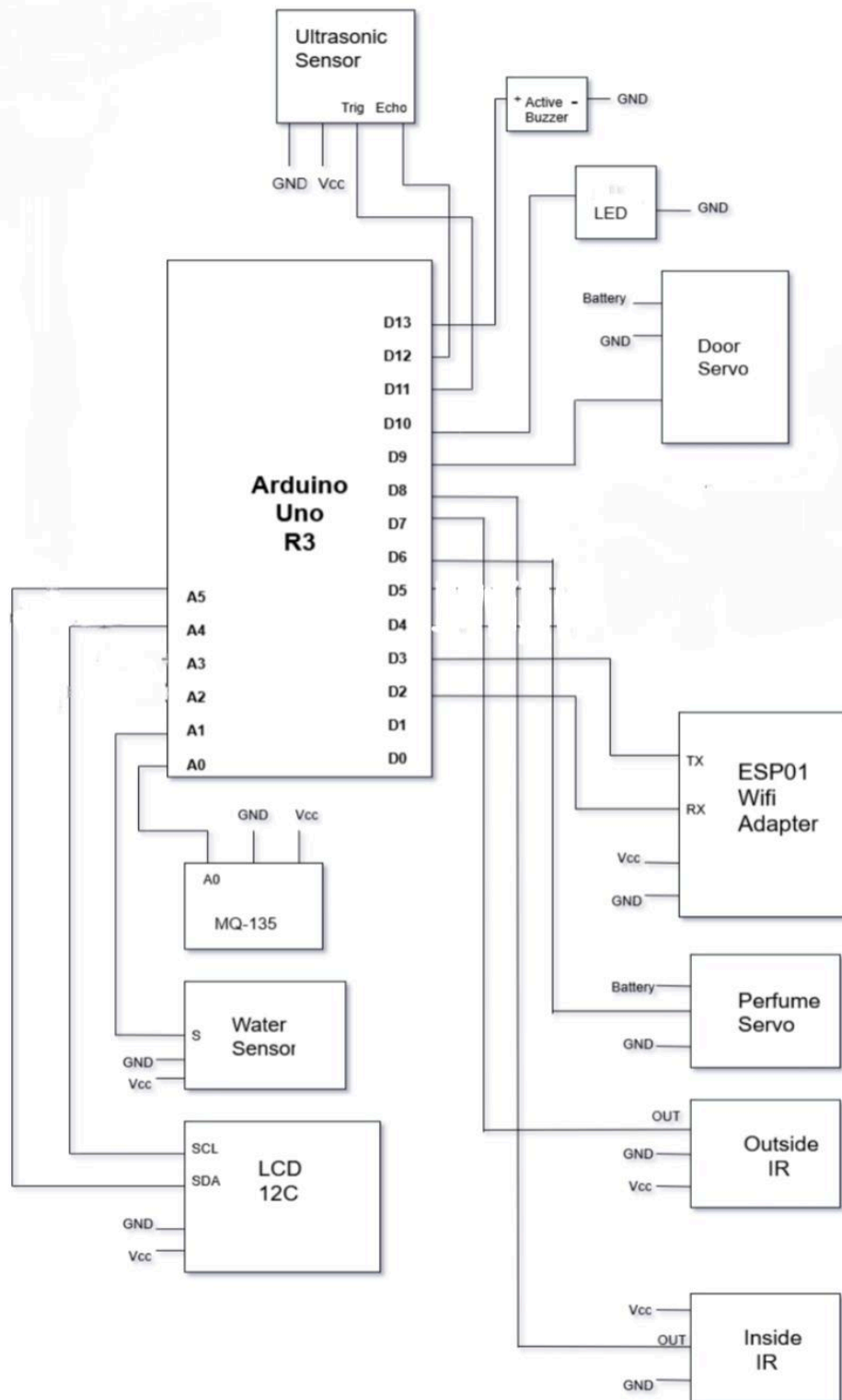
Water Level Sensor	1	50	50
HX711	1	130	130
Load Cell	1	280	280
SG90 Servo Motor	1	135	135
16×2 I2C LCD	1	340	340
LEDs, Buzzer, Buttons	Lot	100	100
Resistors, Wires	Lot	80	80
4× AA Battery Holder	1	100	100
4× AA Batteries	1	200	200
Others			175

Total Estimated Cost			3000 BDT
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- **Communication Protocol**

1. Arduino and the LCD screen can communicate because of I2C (Inter-Integrated Circuit) protocol. It has only two wires of communication (SDA and SCL), which makes it incredibly efficient and suitable for projects on pin-constrained boards such as the Arduino Uno.
2. Serial (SoftwareSerial through UART): The ESP-01 WiFi module may be commanded on the Arduino via this interface. Data analysis and troubleshooting is displayed in the Serial Monitor of the Arduino IDE.

Circuit/Schematic



Implementation:

1. Starting the project/Initialization:
 - All sensors, actuators, LCD and communication modules are started with Arduino Uno.
 - Calibration of the load cell is carried out.
 - The connection to WiFi through ESP-01 module is achieved.
2. User Input Handling:

It has functions such as LCD navigation and system testing.
3. Occupancy Detection:
 - HC-SR04 ultrasonic sensor is used to constantly measure the distance to detect user presence.
 - Debounce (1.5-s) software allows a stable collection of entering/exiting users.
4. Door Automation
 - There are two IR sensors (indoor and out) that monitor entry and exit.
 - The door opens/closes automatically to be operated touchlessly using a servo motor (SG90).
5. Environmental Monitoring and Resource Monitoring:
 - Air quality monitors using MQ-135 gas sensor - activates perfume dispenser (MG996R servo).
 - Water sensor detects floods - sends alert.
 - Occupants load cell measures toilet paper levels - low-paper alarm.
6. IoT Alerting:
 - ESP-01 WiFi module transmits the data to ThingSpeak through the HTTP requests.
 - ThingSpeak React applications send notifications to the Telegram to the maintenance staff.

Challenges:

1. Servo Power Instability: High current draw led to voltage dips. Problem solved with a separate battery supply and software-based motion scaling.
2. ESP-01 Firmware Issues were resolved by manually flashing the right AT firmware and using ThingSpeak as an HTTP intermediate.
3. To address sensor fluctuation, software debugging was used to eliminate invalid or false readings.

4. Sustainability & Impact

- **Sustainability:** The project is mainly created for the purpose of making a cost and energy efficient automated restroom management system. This project has an impact on sustainable resources and management systems. Such as:
 1. It's energy efficient because when the servo motor is idle it can be detached to save power that reduces power consumption.
 2. The automated and controlled flashing system could prevent access water usages which will stop unnecessary water usage and save water.
 3. The toilet paper monitoring will reduce waste and provide a solid waste recycling system.
 4. The flood detection will prevent water damage or unhygienic conditions.
 5. Touchless systems make it more diseases or bacteria transmission-proof.

- **Impact:** This project has many societal, industrial, or community benefits. Such as:
 1. Maintenance staff receive notification or alert to prevent any unwanted situation.
 2. It's low cost, so this is easily implementable.
 3. Flood prevention and automated door systems make it more hygienic.

- **Future Work:**
 1. Multi-stall expansion with I/O expanders
 2. Web-based monitoring dashboard.
 3. Automated flushing system.

- **Limitations:**
 1. Power supply: The system's servos are supplied by an external 4×AA battery pack, suitable for demonstration but not ideal for long-term usage. For continuous, dependable use, a permanently wired power supply is necessary.
 2. Network Dependency: The IoT alerting system is fully dependent on reliable WiFi connectivity. In the event of a network outage, remote alerts via ThingSpeak and Telegram would fail, restricting the proactive maintenance feature.

5. Results & Discussion

The project effectively met all of its key objectives.

1. The occupancy system's 1.5-second debounce timer prevented state flickering, allowing for reliable detection of user entry and exit from the stall.

2. The IR sensors controlled the door mechanism flawlessly, opening for entry when vacant and exiting when occupied.
3. The IoT alarm system was tested for all three conditions: flood, low paper, and service. Each test resulted in a successful notice to the target Telegram account within 5-10 seconds of the Arduino event triggering.
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5. The LCD display properly shows the output

6. Conclusion

This project was able to design and implement an Automated Restroom Management System which improves the hygiene, efficiency and responsiveness in the maintenance of a university environment. The system is able to offer solutions to the actual challenges of sanitation by a sustainable and scalable solution through a combination of sensor, actuators, and IoT services via efficient computer interfacing. The project raises the prospect of embedded systems as a way to establish a material impact on society, at the same time keeping it affordable and sensible to implement it in the general infrastructure of the population.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Hironmoy Sikder		Software design, Circuit building and Management.	- Wrote the C code for the Arduino project - Implemented the ThingSpeak website and telegram. - Build the Environmental Monitoring and Designed the model of the project.
Ummay Maimona Chaman		Hardware Design, Writing.	- Implemented the buzzer functionalities . - Implemented the Automated door system. - Report writing.
Labib Hasan		Hardware Design, Project Modelling.	- Set up the automatic perfume logic. - Water sensor management. - Organized all the necessary components for the project.

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